

Week 8: Combinatorial Foundation

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※ Divisibility

Determine whether each statement is true or false.

- 7 divides 84
- 8 divides 90
- 12 divides 144
- 15 divides 105
- 9 divides 1,001

In general, we say that a divides b if b can be written as $a \times k$ for some whole number k . For example,

$$84 = 7 \times 12,$$

so 7 divides 84. However,

$$90 \div 8 = 11.25,$$

which is not a whole number, so 8 does not divide 90.

1.1 Common Divisibility Rules

2	Last digit is even
3	Sum of digits is divisible by 3
4	Last two digits divisible by 4
5	Ends in 0 or 5
6	Divisible by 2 and 3
8	Last three digits divisible by 8
9	Sum of digits divisible by 9
10	Ends in 0
11	Alternating digit sum is divisible by 11

With this, we can easily test really large numbers and see if they are divisible by some number.

Is 35,244 divisible by 9?

The digit sum is

$$3 + 5 + 2 + 4 + 4 = 18.$$

Since 18 is divisible by 9, then 35,244 is divisible by 9.

※ Primes

A **prime** number has exactly two positive divisors: 1 and itself. Examples include

$$2, 3, 5, 7, 11, 13, 17, \dots$$

A **composite** number, on the other hand, has more than two divisors. Examples include

$$4, 6, 8, 9, 10, 12, \dots$$

Noticing primes and super important. So, it is very crucial you get very comfortable determining whether a number is prime or not.

To determine whether n is prime, it is enough to test divisibility by primes up to $\sqrt{n}$.

Is 97 prime?

Since $\sqrt{97} < 10$, then we only need to test

$$2, 3, 5, 7$$

as these are the only primes less than 10. You can check none divide 97, so 97 is prime.

Here are some random, but very common facts:

- 2 is the only even prime.
- Every integer greater than 1 is either prime or composite.
- There are infinitely many primes.

※ Prime Factorization

Write each number as a product of smaller numbers.

$$12, \quad 18, \quad 30, \quad 72$$

Can you continue breaking them apart until only prime numbers remain?

There is a tedious strategy to finding the prime factorization of a number.

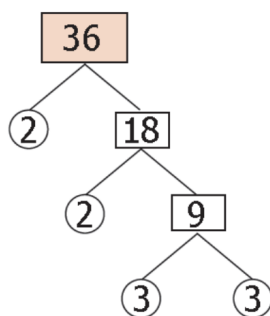
1. If you have a number N , go through all the prime numbers up until $\sqrt{N}$.
2. If you found a prime number p and it divides N , then write down the prime number p and go back to the first step with $\frac{N}{p}$.
3. If you eventually cannot find a prime less than $\sqrt{N}$, then the number you have is prime since it has no divisors! Write down N since it is prime, then go to the last step.
4. Once you arrive at 1 you are done.

We will try this on 30 together:

- The first prime we know of is 2. It does divide 30! This gives us a prime 2 and a new number $\frac{30}{2} = 15$.
- Now, with a number 15, we will go up to all the prime numbers from 1 to $\sqrt{15} < 4$. We start with the first prime 2. It doesn't divide 15. Then, 3 is our next prime. It does divide 15! So, 3 is our next prime. We go to the next step with 5.
- Now, with the number 5, we will go up to all the primes up to $\sqrt{5} < 3$. We start with the first prime 2, which doesn't divide 5. Our next prime 3 is already greater than $\sqrt{5}$. This means 5 is prime!
- Now, we arrived at 1 and we are done.

Nicely enough, our written down numbers are 2, 3, and 5. So, $30 = 2 \cdot 3 \cdot 5$.

In fact, it is very common for people to make “factor trees” when finding the prime factorization of such numbers. As you do more problems, you will easily be able to recognize primes instantly and figure out



prime factorizations much easier. One piece of information that might help is that: every number has a unique prime factorization. So, there will almost never be any confusion. I suggest trying it on 12, 18, and 72. You should get the following, but try and use the same process and see if we get the same result!

$$12 = 2^2 \cdot 3$$

$$18 = 2 \cdot 3^2$$

$$72 = 2^3 \cdot 3^2$$

※ Greatest Common Divisor

Find the largest number that divides both:

- 12 and 18.
- 24 and 36.
- 30 and 45.

The **Greatest Common Divisor (GCD)** of two numbers is the largest positive integer that divides both of them. We denote the Greatest Common Divisor of a and b as $\gcd(a, b)$.

Nicely enough, there is a much faster way than writing out the divisors of a number and seeing what the largest common value is.

The way actually involves using the prime factorization. Imagine two numbers a and b . Suppose they have the following prime factorizations:

$$\begin{aligned} a &= 2^5 \cdot 3^6 \cdot 5^2 \cdot 7^3 \\ b &= 2^9 \cdot 3^8 \cdot 5^1 \cdot 7^5 \end{aligned}$$

Does a divide b ? Well, let's recall a divides b if $\frac{b}{a}$ is an integer. In this example,

$$\frac{2^9 \cdot 3^8 \cdot 5^1 \cdot 7^5}{2^5 \cdot 3^6 \cdot 5^2 \cdot 7^3} = \frac{2^4 \cdot 3^2 \cdot 7^2}{5}$$

Notice, 5 cannot divide any of the values in the numerator. That is because primes are the most basic unit. From dividing out the prime factorizations, 5 remains in the denominator and doesn't cancel out entirely. So, in general, this means that for a number to divide another number, the exponent in each corresponding value of a must be less than that of b . For instance, if instead, we had

$$\begin{aligned} a &= 2^5 \cdot 3^6 \cdot 5^2 \cdot 7^3, \\ b &= 2^9 \cdot 3^8 \cdot 5^2 \cdot 7^5. \end{aligned}$$

Let's compare the exponents. For the prime 2, $5 \leq 9$. For 3, $6 \leq 8$. For 5, $2 \leq 2$. For 7, $3 \leq 5$. This means a divides b since the exponents are less than or equal to. In the prior example the exponent of 5 in a is greater than that of b so a doesn't divide b .

All of this helps us motivate the following idea: If we have two numbers with the prime factorizations

$$\begin{aligned} a &= p_1^{a_1} \cdot p_2^{a_2} \cdot p_3^{a_3} \cdot p_4^{a_4} \cdots, \text{ and} \\ b &= p_1^{b_1} \cdot p_2^{b_2} \cdot p_3^{b_3} \cdot p_4^{b_4} \cdots, \end{aligned}$$

then

$$\gcd(a, b) = p_1^{\min(a_1, b_1)} \cdot p_2^{\min(a_2, b_2)} \cdot p_3^{\min(a_3, b_3)} \cdot p_4^{\min(a_4, b_4)} \cdots$$

We will use this example on 30 and 45 from the previous example.

- The prime factorization of 30 and 45 is $2^1 \cdot 3^1 \cdot 5^1$ and $2^0 \cdot 3^2 \cdot 5^1$.
- So, the greatest common divisor is $2^{\min(1,0)} \cdot 3^{\min(1,2)} \cdot 5^{\min(1,1)} = 2^0 \cdot 3^1 \cdot 5^1 = 15$.

※ Least Common Multiple

Find the smallest number is divisible by both:

- 12 and 18.
- 24 and 36.
- 30 and 45.

The **Least Common Multiple (LCM)** of two numbers is the smallest positive integer that is divisible by both of them. We denote the Least Common Multiple of a and b as $\text{lcm}(a, b)$.

Using similar ideas from above: If we have two numbers with the prime factorizations

$$a = p_1^{a_1} \cdot p_2^{a_2} \cdot p_3^{a_3} \cdot p_4^{a_4} \cdots, \text{ and}$$

$$b = p_1^{b_1} \cdot p_2^{b_2} \cdot p_3^{b_3} \cdot p_4^{b_4} \cdots,$$

then

$$\text{lcm}(a, b) = p_1^{\max(a_1, b_1)} \cdot p_2^{\max(a_2, b_2)} \cdot p_3^{\max(a_3, b_3)} \cdot p_4^{\max(a_4, b_4)} \cdots.$$

We will use this example on 30 and 45 from the previous example.

- The prime factorization of 30 and 45 is $2^1 \cdot 3^1 \cdot 5^1$ and $2^0 \cdot 3^2 \cdot 5^1$.
- So, the greatest common divisor is $2^{\max(1,0)} \cdot 3^{\max(1,2)} \cdot 5^{\max(1,1)} = 2^1 \cdot 3^2 \cdot 5^1 = 90$.

※ Challenge Examples

There will never be a problem that asks you: “Is X prime?”, “What is the LCM of X and Y ?”, or “What is the prime factorization of X ?”. Instead, problems will ask deeper questions. We will see some here as well as in the “practice problems”.

The 5-digit number 2018U is divisible by 9. What is the remainder when this number is divided by 8?

Recall, for 2018U to be divisible by 9, then

$$2 + 0 + 1 + 8 + U = 11 + U$$

is divisible by 9. This implies $U = 7$.

So, the number is 20187. We want the remainder when divided by 8. Recall, from our rules, this is equivalent to the remainder when dividing the last three digits of 187 by 8.

Note that $187 = 8 \cdot 23 + 3$, so the remainder is 3.

Janelle writes the positive integers less than or equal to 26 on a white board. What is the sum of the multiples of 3 that she wrote?

Note that the integers less than or equal to 26 but are also multiples of 3 include:

$$3, 6, 9, 12, 15, 18, 21, 24$$

Pairing up $3 + 24$, $6 + 21$, $9 + 18$, and $12 + 15$, then there are four pairs each summing to 27. So, $4 \cdot 27 =$ 108.

※ Practice Problems

- Find the sum of all positive multiples of 3 that are factors of 27.
- What is the value of $\gcd(1, 9) \times \gcd(2, 8) \times \gcd(3, 7) \times \cdots \times \gcd(9, 1)$?
- At 5 : 57, all the digits on the clock are prime numbers. In how many minutes will all the digits on the clock be prime numbers again?
- What is the greatest perfect square that evenly divides both 2026^2 and 2^{2026} ?
- Suppose that the greatest common divisor of n and 5040 is equal to 120. Determine the sum of the four smallest possible positive integers n .
- Let N be a two-digit integer. If the sum of the digits of N is a multiple of 6 and the product of the digits of N is a multiple of 7, what is the sum of all possible values of N ?
- If n is an integer for which the greatest common factor of n and 18 is 6 and the least common multiple of n and 9 is 126, what is the value of n ?
- The sum of two prime numbers is 85. What is the product of these two prime numbers?
- The 7-digit numbers $\underline{7} \underline{4} \underline{A} \underline{5} \underline{2} \underline{B} \underline{1}$ and $\underline{3} \underline{2} \underline{6} \underline{A} \underline{B} \underline{4} \underline{C}$ are each multiples of 3. Which of the following could be the value of C ?
- Wurmples wrote the following message:

“My favorite number is the sum of all eight two-digit multiples of ■.”

But the last number got smudged! What is Wurmples’s favorite number?

- The digits 1, 2, 3, 4, and 5 are each used once to write a five-digit number $PQRST$. The three-digit number PQR is divisible by 4, the three-digit number QRS is divisible by 5, and the three-digit number RST is divisible by 3. What is P ?
- Show that $\gcd(a, b) \cdot \text{lcm}(a, b) = ab$.